

YIWU, China

WMO: 521180

Lat: 43.2422N

Lon: 94.6724E

Elev: 1729

StdP: 82.21

Time Zone: 8.00 (E08)

Period: 94-02

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -21.3	(c) -19.4	(d) -30.7	(e) 0.3	(f) -17.5	(g) -28.7	(h) 0.3	(i) -14.5	(j) 13.0	(k) -4.1	(l) 10.9	(m) -11.9	(n) 2.4	(o) 270	(p) 0.623

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.0	(c) 27.5	(d) 14.1	(e) 26.4	(f) 13.6	(g) 25.2	(h) 13.0	(i) 16.1	(j) 23.9	(k) 15.1	(l) 23.6	(m) 14.2	(n) 22.7	(o) 2.9	(p) 40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.1	11.6	19.7	11.8	10.6	19.1	10.6	9.8	18.2	51.7	23.7	48.3	23.7	45.8	22.5	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.5	9.5	8.0	-24.3	29.7	2.3	1.0	-25.9	30.4	-27.3	31.0	-28.6	31.5	-30.2	32.3
			-24.7	18.5	2.2	2.7	-26.2	20.5	-27.5	22.1	-28.8	23.6	-30.4	25.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.9	-11.0	-7.3	-2.5	5.8	13.0	17.1	19.1	17.8	12.6	4.5	-2.5	-8.8		
	DBStd	11.35	4.09	3.84	5.61	4.53	4.07	3.05	2.37	2.89	3.83	4.71	5.10	4.62		
	HDD10.0	2824	650	483	386	136	19	1	0	1	16	177	374	582		
	HDD18.3	5004	908	717	644	376	169	57	20	43	176	430	623	840		
	CDD10.0	957	0	0	0	10	112	213	281	243	93	6	0	0		
	CDD18.3	96	0	0	0	0	4	20	42	26	3	0	0	0		
	CDH23.3	782	0	0	0	2	47	159	304	235	34	0	0	0		
	CDH26.7	73	0	0	0	0	2	14	34	21	2	0	0	0		
(14) Wind	WSAvg	3.3	2.6	2.7	3.3	4.3	4.5	3.8	3.4	3.2	3.3	3.0	2.7	2.6		
(15) Precipitation	PrecAvg	90	1	1	2	5	8	19	27	14	8	3	1	1		
	PrecMax	152	3	3	18	28	29	55	78	28	28	10	7	5		
	PrecMin	35	0	0	0	0	0	2	4	0	0	0	0	0		
	PrecStd	30	1	1	3	5	7	11	19	9	8	3	1	1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.3	5.5	15.6	22.6	26.7	28.2	29.2	28.5	26.2	19.0	12.7	4.9		
		MCWB	-1.5	-1.5	4.2	8.3	11.1	13.6	15.6	14.5	12.0	7.6	3.2	-1.5		
	2%	DB	0.0	2.9	11.2	18.0	24.6	26.6	27.5	27.2	24.0	16.3	9.3	2.1		
		MCWB	-4.3	-3.0	2.6	6.4	10.0	12.7	15.3	13.9	10.6	6.1	1.4	-3.2		
	5%	DB	-2.5	1.1	8.9	15.7	22.8	25.1	26.4	26.0	21.9	14.3	7.2	0.0		
		MCWB	-6.0	-4.1	0.8	5.3	9.2	12.1	14.7	13.3	10.1	5.0	0.1	-4.4		
	10%	DB	-4.7	-0.6	6.7	13.7	20.8	23.9	25.1	24.6	19.9	12.6	5.1	-2.1		
		MCWB	-7.6	-5.1	-0.2	4.5	8.8	11.8	14.3	12.7	9.2	4.4	-0.8	-5.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.5	-1.2	4.7	8.5	12.2	15.2	17.9	17.0	13.1	8.9	3.5	-1.3		
		MCDB	3.1	4.3	13.6	20.8	24.5	25.0	26.1	24.5	23.2	17.7	12.1	4.3		
	2%	WB	-4.1	-2.6	2.9	6.9	11.1	13.9	16.7	15.1	11.9	6.6	1.6	-2.9		
		MCDB	-0.4	2.2	10.3	16.2	22.6	23.1	24.6	23.5	21.5	14.8	8.8	1.8		
	5%	WB	-6.0	-3.6	1.2	5.9	10.1	13.2	15.7	14.3	10.7	5.4	0.5	-4.4		
		MCDB	-2.8	0.9	8.3	14.4	21.1	22.4	23.8	23.2	20.4	13.1	6.6	-0.3		
	10%	WB	-7.6	-5.0	-0.1	4.9	9.1	12.7	15.0	13.6	9.6	4.5	-0.6	-5.8		
		MCDB	-4.8	-0.7	6.3	12.4	19.2	21.8	23.4	22.6	18.9	11.8	4.9	-2.3		
(35) Mean Daily Temperature Range	MDBR		10.0	10.6	10.9	10.2	10.5	10.0	10.0	10.7	11.2	10.4	9.8	9.9		
	5% DB	MCDBR	11.5	11.7	13.0	12.9	12.8	11.9	11.8	12.6	13.1	12.3	11.6	11.6		
		MCWBR	8.4	7.8	7.4	6.3	6.0	5.8	5.9	5.4	5.8	6.5	6.9	8.1		
	5% WB	MCDBR	11.3	10.7	11.7	10.4	10.4	9.7	10.2	10.7	11.5	11.6	11.0	11.2		
(40) Clear-Sky Solar Irradiance	taub		0.276	0.316	0.360	0.407	0.418	0.430	0.424	0.395	0.364	0.337	0.299	0.274		
	taud		2.290	2.203	2.089	2.022	2.050	2.097	2.155	2.226	2.253	2.238	2.290	2.304		
	Ebn at Noon		856	870	869	855	855	844	845	858	858	832	815	822		
	Edh at Noon		84	110	141	164	164	157	147	132	119	105	84	76		
(44) All-Sky Solar Radiation	RadAvg		2.24	3.31	4.75	5.96	6.94	6.88	6.74	6.24	5.30	3.88	2.51	1.92		
	RadStd		0.12	0.17	0.21	0.33	0.34	0.45	0.32	0.25	0.21	0.14	0.15	0.14		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	+0.46	N/A	N/A	+0.65	-1.02	-1.46	N/A	N/A	+146	+101

Nomenclature: See separate page